

Supporting Information

PD-L1 glycosylation and its impact on binding to clinical antibodies

Julius Benicky^{1,3,#}, Miloslav Sanda^{1,3,#}, Zuzana Brnakova Kennedy^{1,3}, Oliver C. Grant⁴, Robert J. Woods⁴, Alan Zwart¹, Radoslav Goldman^{1,2,3,}*

¹ Department of Oncology, Lombardi Comprehensive Cancer Center, Georgetown University, Washington, D.C. 20057, ² Department of Biochemistry and Molecular & Cellular Biology, Georgetown University, Washington, DC, 20057, ³ Clinical and Translational Glycoscience Research Center, Georgetown University, Washington, DC, 20057, ⁴ Department of Biochemistry and Molecular Biology, Complex Carbohydrate Research Center, University of Georgia, Athens, GA 30602

Ligand	Analyte	K_D (nM)	k_a (10^4 /Ms)	k_d (10^{-4} /s)
Avelumab	gPD-L1	0.4	6.74	0.29
	dgPD-L1	17.0	0.23	0.39
Durvalumab	gPD-L1	0.2	73.73	1.73
	dgPD-L1	0.4	48.53	1.86
Atezolizumab	gPD-L1	4.7	23.57	10.98
	dgPD-L1	17.8	6.87	12.25

Supplemental Table S1. Kinetic parameters of interaction between antibodies and glycosylated (gPD-L1) or deglycosylated PD-L1 (dgPD-L1) analyzed by SPR assay with immobilized PD-L1 antibodies. K_D – dissociation constant, k_a – association rate constant, k_d – dissociation rate constant.

Site	3 major structures	Before			After		
		SO	GP#	%	SO	GP#	%
N35	HexNAc(6)Hex(4)SA(1)	>99%	44	13	>99%	73	14
	HexNAc(6)Hex(5)			8			5
	HexNAc(6)Hex(4)Fuc(1)			6			8
N192	HexNAc(4)Hex(4)	100%	75	15	>99%	60	13
	HexNAc(4)Hex(4)Fuc(1)			13			12
	HexNAc(4)Hex(5)			12			11
N200	HexNAc(5)Hex(4)Fuc(1)SA(2)	81%	61	19	76%	38	20
	HexNAc(5)Hex(4)SA(2)			18			18
	HexNAc(5)Hex(4)Fuc(2)			5			2
N219	---	<0.1%	NQ	NQ	<0.1%	NQ	NQ

Supplemental Table S2. Glycosylation of the secreted form of PD-L1 (Phe19-Thr239) produced in HEK293 cells and analyzed on four glycopeptides before and after immunoaffinity purification. The results show top 3 structures identified on PD-L1 before/after purification. Abbreviations: HexNAc, N-acetylhexosamine; Hex, Hexose; Fuc, Fucose; SA, sialic acid; GP#, number of identified glycopeptides; NQ, not quantifiable; SO, site occupancy; GP#, number of identified glycopeptides

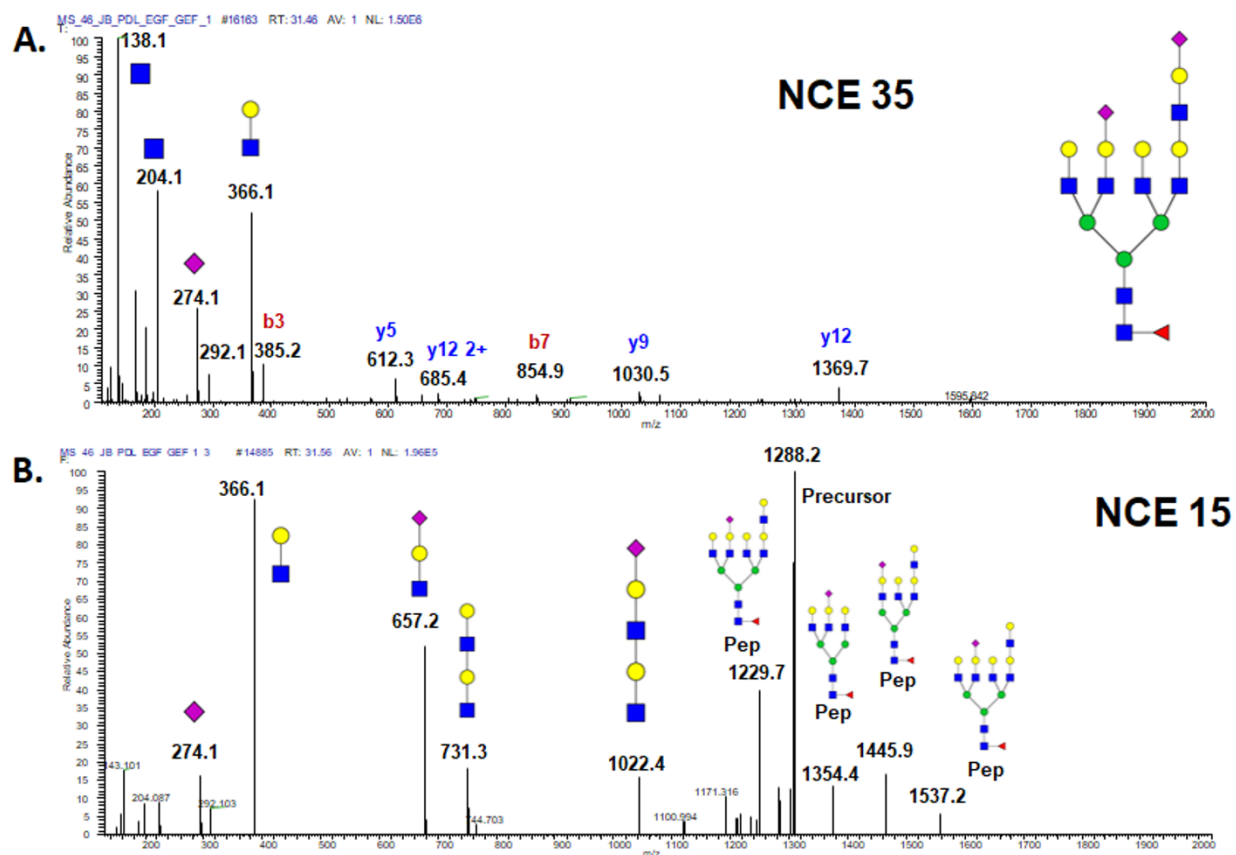


Figure S1. High (NCE 35) and low collision (NCE15) HCD tandem mass spectra of PD-L1 glycopeptide N219 produced in MDA-MB-231 cells. Series of peptide backbone fragment ions confirm occupied glycopeptide N219 in high collision energy spectra (**A**) and polyLacNAc-specific oxonium ions confirm the presence of polyLacNAc outer arm structure motif (**B**).